Sustainable Cities and Society: Advances 2 (2026) 100069

Contents lists available at ScienceDirect

Sustainable Cities and Society: Advances

journal homepage: www.sciencedirect.com/journal/sustainable-cities-and-society-advances

Citizen-driven monitoring of sustainable diets: Linking individual behaviour to territorial indicators

Patrícia Abrantes *, Eduarda Marques da Costa

Centro de Estudos Geográficos (CEG), Instituto de Geografia e Ordenamento do Território (IGOT), Universidade de Lisboa, Rua Branca Edmée Marques, 1600-276 Lisboa, Portugal

A R T I C L E I N F O

Keywords:
Sustainable Diet Score
Geographic citizen science
Monitoring
Perception-behaviour gap
Portugal

A B S T R A C T

Cities, metropolitan areas and municipalities increasingly carry responsibility for promoting sustainable diets through public procurement, health promotion, and diet-related local food policies, yet they often lack low-burden indicators that capture everyday dietary behaviours and citizens' perceptions in a way that supports routine policy monitoring. This study develops and tests Care4Food, a geographic citizen-science, behaviour-based Sustainable Diet Score paired with a self-perceived diet sustainability measure, and demonstrates how citizen-generated data can be translated into municipal-level indicators that can support broader urban and regional sustainability governance in Portugal.

First, a framework-aligned scoring system was constructed and calibrated using a computer-assisted telephone interview survey of 1813 adults in mainland Portugal. The Care4Food score integrates four sub-domains normalised to a 1–5 scale: protein balance, fruit and vegetable intake, ultra-processed food intake, and sustainability-oriented practices. The mean Care4Food score was 3.08 (SD = 0.82), with lower values primarily driven by protein imbalance and weaker sustainability practices. Self-perceived diet sustainability was higher (mean = 4.0), producing a perception–behaviour gap and a weak correlation between perceived and behaviour-based measures (r = 0.29). Miscalibration was strongest among low-scoring participants, consistent with a Dunning–Kruger-type pattern. Second, the calibrated score was implemented through a digital survey linked to an interactive dashboard to enable automated scoring, immediate personalised feedback, and aggregation of results at the municipal scale, while visualising these indicators alongside contextual food-environment layers. Forty voluntary online submissions generated descriptive exploratory municipal-level indicators and reproduced the observed tendency toward overestimation. The study demonstrates the feasibility of a low-cost, transparent, citizen-driven monitoring instrument that complements conventional urban sustainability and smart-city monitoring systems by linking individual dietary practices and perceptions to municipal decision-making.

1. Introduction

Food systems are a major source of environmental pressure and a central determinant of population health. They account for roughly one-third-of anthropogenic greenhouse gas emissions across production-to-consumption chains, contribute substantially to land-use change, and use close to 70% of global freshwater withdrawals (FAO & WHO, 2019; Willett et al., 2019). At the same time, unhealthy diets remain among the leading risk factors for premature mortality worldwide, contributing to obesity, cardiovascular disease, diabetes, and micronutrient deficiencies (Afshin et al., 2019). Urbanisation alters food environments by increasing exposure to ultra-processed foods, changing mobility patterns, and reshaping access to fresh products, thereby influencing everyday dietary behaviours. As urbanization accelerates and incomes rise, public policies are under growing pressure to deliver diets that are nutritionally adequate while also environmentally sustainable, affordable, and socially acceptable (Fanzo et al., 2020; FAO & WHO, 2019). Although agrifood systems challenges are global, many of the policy instruments intended to address them, particularly on the food consumption side, operate at the metropolitan, city, and municipal level. Municipalities are increasingly involved in food-system governance and planning through responsibilities such as public food procurement, particularly through school meals and other public catering services, health and well-being promotion, and land-use planning decisions that

* Corresponding author.
E-mail addresses: patricia.abrantes@edu.ulisboa.pt (P. Abrantes), eduarda.costa@edu.ulisboa.pt (E. Marques da Costa).

https://doi.org/10.1016/j.scsadv.2026.100069
Received 11 February 2026; Received in revised form 12 April 2026; Accepted 12 May 2026
Available online 14 May 2026

influence local agrifood systems and dietary patterns (Rocha & Pires, 2025). As a result, urban areas are expected to translate international dietary and sustainability objectives into concrete local actions (Cirone et al., 2023).

International frameworks provide clear normative guidance in this regard. The FAO/WHO Guiding Principles for Sustainable Healthy Diets (FAO & WHO, 2019) and the Planetary Health Diet proposed by the EAT–Lancet Commission (Willett et al., 2019; EAT–Lancet Commission, 2025) are increasingly referenced in urban food strategies and policies. However, achieving food-system objectives consistent with these frameworks requires monitoring tools capable of assessing whether dietary practices and related behaviours are changing in line with national and global sustainability goals (FAO, IFAD, UNICEF, WFP & WHO, 2024; HLPE, 2020), including mechanisms that link system-level goals with individual dietary behaviours (Polman et al., 2025)

Current monitoring approaches remain poorly adapted to these behavioural and place-based monitoring needs. Global tools such as the Food Systems Dashboard, the State of Food Security and Nutrition in the World (SOFI), and the Food Sustainability Index rely mainly on macro-level indicators related to food availability, prices, production, dietary costs, and nutrition outcomes (GAIN et al., 2024; JRC, 2024). These indicators are essential for national and international reporting, but they offer limited insight into everyday dietary behaviours. In particular, they provide little information on how local policies shape what people eat, or on how dietary practices vary across places within countries.

Several research-based indices attempt to address this limitation by measuring sustainable diets at the individual level. Tools such as the Sustainable Diet Index (SDI), the Dietary Pattern Sustainability Index (DIPASI), and the Planetary Health Diet Index (PHDI) combine nutritional, environmental, economic, and sociocultural dimensions using detailed dietary intake data (Cacau et al., 2021; Jung et al., 2024; Neta et al., 2023; Dias et al., 2024; Seconda et al., 2020; Trijsburg et al., 2020). While these approaches are methodologically robust, they depend on resource-intensive data collection and modelling, which limits their usefulness for routine municipal monitoring or ongoing policy processes. Survey-based, behaviour-oriented approaches have emerged as more pragmatic alternatives, but they are rarely embedded in spatial or governance frameworks that make them directly usable for local decision-making (Culliford & Bradbury, 2020; Polzin et al., 2023).

Citizen participation and crowd-sourced data are therefore increasingly promoted to support urban policy and decision-making. In practice, however, their application often remains fragmented or symbolic. In many cases, participatory approaches fail to generate data that can be sustained over time or meaningfully integrated into policy processes (Mohammadi, 2025; Adewopo et al., 2025; Harrington et al., 2021; Manners et al., 2022; Oakden et al., 2021; Seok et al., 2023). In the food-system domain, geographic citizen science and Volunteered Geographic Information (VGI) initiatives have focused mainly on food prices, food waste, or food accessibility, with much less attention to individual dietary behaviours and their aggregation at policy-relevant spatial scales (Arbia et al., 2023; Aylward et al., 2022; Haworth et al., 2016; Levin et al., 2017; Mendoza et al., 2024; Müller et al., 2024). As a result, the potential of low-cost, citizen-generated data to support continuous, place-based monitoring of sustainable diets remains largely underutilised.

Taken together, municipalities face a persistent dilemma. They are increasingly expected to act on sustainable agrifood systems, yet often lack monitoring instruments that are affordable, behaviourally meaningful, and aligned with the scale at which local decisions are made. In particular, data that capture everyday dietary behaviours and relate them to municipal responsibilities remain scarce. This limits the ability of local authorities to benchmark performance, identify priority areas, or assess whether interventions are producing meaningful change.

This study addresses these gaps by developing the Care4Food Sustainable Diet Score (hereafter, Care4Food Score), a geographic, citizen-driven, behaviour-based indicator of dietary sustainability, combined with a self-perceived sustainability measure and spatialised at the municipal level.

First, the score is constructed and calibrated using data from a large-scale survey of adults in mainland Portugal (n = 1813), drawing on established sustainable diet frameworks and examining discrepancies between perceived and behaviour-based sustainability. Evidence from behavioural research suggests that perceived and actual diet quality are often poorly aligned, with individuals frequently overestimating the healthfulness or sustainability of their eating habits (Dickson-Spillmann & Siegrist, 2011; de Ridder et al., 2012). If such perception–behaviour gaps are not considered, information-based or awareness-focused interventions may have limited effect, particularly among those whose diets are least aligned with sustainability objectives (Dijksterhuis & Wallner, 2025; Thøgersen, 2025).

Second, the calibrated score is implemented through a low-cost digital survey and geospatial pipeline that enables continuous data collection, immediate feedback to participants, and aggregation of individual responses into municipal-level indicators.

The study follows three objectives:

(1) to construct and calibrate a framework-aligned sustainable diet score based on self-reported dietary behaviours;
(2) to quantify perception–behaviour gaps in dietary sustainability; and
(3) to demonstrate how citizen-generated data can support low-cost, municipal-level monitoring for public policy.

By embedding simplified behavioural indicators within a spatial data infrastructure, this study contributes a practical tool to support municipal-level decision-making. It provides policy-relevant monitoring indicators that bridge food-system science, nutrition, and spatial planning in a systemic way, and responds directly to calls from the EU Farm to Fork Strategy for improved monitoring of dietary transitions and diet-related dimensions of food-system change at sub-national levels, as well as national strategies for healthy and sustainable food systems (HLPE, 2020; European Commission, 2020).

2. Data and methods

2.1. Study area

The study was conducted in mainland Portugal, excluding the autonomous regions of the Azores and Madeira due to their insular characteristics, which differ substantially from the mainland in terms of food systems, geography, and socio-economic context.

For the baseline survey, analyses were conducted using the Nomenclature of Territorial Units for Statistics (NUTS) II as the primary spatial unit, comprising seven mainland regions: Norte, Centro, Grande Lisboa, Península de Setúbal, Oeste e Vale do Tejo, Alentejo, and Algarve. For the online deployment of the scoring system, individual responses were aggregated at the municipal level, covering all 278 municipalities in mainland Portugal, to support fine-grained spatial monitoring and visualisation of results (Fig. 1).

2.2. Methodology

This study develops a geographic citizen-science to measure sustainable dietary behaviours and perceptions in Portugal. The methodology combines survey-based behavioural scoring, self-perceived diet assessment, and spatial aggregation of responses through a geospatial data pipeline.

The methodological design comprises two interlinked components:

(i) Scoring system development and calibration:
A behaviour-based scoring algorithm grounded in international and national dietary sustainability frameworks converts

Fig. 1. Mainland Portugal NUTSII and municipalities.
Source: DGT, SNIG, 2024.

Table 1
Framework for constructing the Care4Food score in alignment with international guidelines.

Sub-score	Survey variables	Frequency of behaviours - rewarded (high score)	Frequency of behaviours - penalised (low score)	Alignment with international frameworks
Plant and animal protein balance (Score$_{protein}$)	*carne_vermelha* (red meat), *carne_branca* (white meat), *leite* (dairy), *cereais* (cereals), *leguminosas* (legumes)	• Red meat ≤1×/week	• Red meat several times/ week or daily	FAO/WHO (2019): encourage diverse protein sources limit red and processed meat. PHDI (Planetary Health Diet Index) (Cacau et al., 2021): red meat as key constraint.
		• White meat: fish/poultry several times/week	• Lack of balance between plant and animal protein sources	EAT–Lancet (2019, 2025): red meat ≤1/ week; plant proteins as the dietary core and moderate fish/poultry.
		• Moderate dairy consumption	• Bread-only or milk-only dietary patterns (diversity safeguard)	SHED Index (Tepper et al., 2021): includes dairy as balanced option. FAO/WHO (2019) Dairy moderate according to adequacy & cultural fit
		• Daily cereals and/or legumes with diversity across protein sources		Portuguese Food Wheel (DGAS, 2016) : cereals/ legumes as staples; animal protein in moderate amounts; FAO/WHO (2019): encourage diverse protein sources
Fruits & vegetables intake (Score$_{fruits}$)	*vegetais* (vegetables), *frutas* (fruits)	• Daily or several-times-daily fruit and vegetable intake	• Both consumed ≤ once/week	FAO/WHO (2019): multiple daily servings of fruit and vegetable; emphasis on dietary diversity.
		• One consumed daily and the other several times/week	• Strong imbalance (e.g., daily fruit but rare vegetables)	Portuguese Food Wheel (DGAS, 2016): 3–5 servings/day of fruit and 3–5 servings/day of vegetables.
			• Rare or no fruit/ vegetable intake	EAT–Lancet (Willett et al., 2019): ~300 g vegetables + 200 g fruit/day.
Ultra-processed foods intake (UPFs) (Score$_{processed}$)	*snacks, ultraprocessados* (UPFs)	• Never or rarely consume UPFs (≤ monthly)	• Weekly or daily UPF consumption	FAO/WHO (2019): limit foods high in salt, sugar, and unhealthy fats.
		• Avoidance of snacks, sweets, processed meats, and ready meals	• Several-times-daily intake (worst case)	EAT-Lancet (Willett et al., 2019): favors home-prepared, minimally processed meals; PNPAS 2022–2030: reducing fast-food and UPF reliance (DGS (Direção Geral da Saúde), 2022); PHDI (Cacau et al., 2021): UPF limits; NOVA classification (Monteiro et al., 2019): avoid UPFs
Sustainability practices (Score$_{sustentabilidade}$)	*take_away, jantar_fora* (eating out), *onde_compra* (where you buy), *como_desloca* (which transports), *sobras* (waste), *compra_frutas* (fruits and veg buying frequency), *perto_longe* (additional trips)	• Rare/never takeaway or eating out	• Frequent takeaway/ restaurant meals	EAT–Lancet (Willett et al., 2019): emphasis on home-prepared meals and lower-impact food chains; PNPAS 2022–2030 promotes Mediterranean behaviors of home cooking, local fresh food purchasing (DGS

(continued on next page)

Table 1 (*continued*)

Sub-score	Survey variables	Frequency of behaviours - rewarded (high score)	Frequency of behaviours - penalised (low score)	Alignment with international frameworks
				(Direção Geral da Saúde), 2022)
		• Buying from local markets or directly from producers	• Reliance on supermarkets/ hypermarkets	FAO/WHO (2019): preference for seasonal/local foods; reduction of food waste; PNPAS 2022–2030 promotes Mediterranean behaviors of local fresh food purchasing; EU Farm to Fork (2020): shorter supply chains,
		• Walking/cycling or public transport to food outlets	• Systematic use of private car for food shopping	EU Farm to Fork (2020): sustainable mobility. FAO/WHO (2019): environmental impact principle; Neta et al. (2023). this review found transport missing in current indices, and proposes to extend sustainability to mobility and transports within low-impact diets
		• Rare or no food waste • Frequent purchase of fresh produce	• Frequent food waste	HSDI (Harray et al., 2022), includes waste; SHED Index (Tepper et al., 2021); EU Farm to Fork (2020): reduction of food waste.
		• Choosing nearby prod ucts;integrating food shopping into daily routines	• Infrequent purchase of fresh produce	FAO/WHO (2019): preference for seasonal/local foods; PNPAS 2022–2030 promotes Mediterranean behaviors of local fresh food purchasing (DGS (Direção Geral da Saúde), 2022); EU Farm to Fork (2020): shorter supply chains,
			• Additional car trips made solely to buy fresh foods	EU Farm to Fork (2020): sustainable mobility. FAO/WHO (2019): environmental impact principle. Neta et al. (2023).

individual-level survey responses into four sub-scores, composite Care4Food Sustainable Diet Score (hereafter, Care4Food score), and a self-perceived diet sustainability measure. The scoring system was developed and calibrated using data from a large-scale baseline survey, ensuring conceptual alignment with established frameworks and internal consistency across dietary dimensions.

(ii) Digital deployment and spatial aggregation:

The calibrated scoring system was implemented in a web-based geospatial survey platform with integrated dashboard (ArcGIS Survey123 and ArcGIS Online), enabling automatic score calculation at the time of submission and live aggregation of citizen-generated data at the municipal level. This infrastructure supports real-time visualisation of aggregated indicators through an interactive dashboard, facilitating exploratory spatial monitoring of dietary sustainability patterns.

Together, these components allow a calibrated survey-based score to be extended into a dynamic, spatially explicit monitoring prototype, while preserving a stable scoring logic across both the baseline survey and the online deployment.

2.3. Constructing the scoring system through the large-scale baseline survey

In a first step, a structured survey was designed to operationalise a behaviour-based sustainable diet score, drawing on major international and national sustainable diet frameworks. These included the FAO/WHO Guiding Principles for Sustainable Healthy Diets (2019), the EAT–Lancet recommendations (Willet et al., 2019), the Portuguese Food Wheel together with the Mediterranean dietary guidance (DGS, 2024, 2016). The survey was designed to capture dietary behaviours across food groups, food-security experiences, sustainability-related practices, self-perceived diet sustainability, and socio-demographic characteristics.

Data was collected from a sample of 1813 adults (≥18 years) using computer-assisted telephone interviewing (CATI). A quota sampling strategy was applied based on gender, age group (18–24, 25–44, 45–64, ≥65), and NUTS II region, ensuring a balanced sample with an estimated margin of error of approximately ±2% at the 95% confidence level. Respondents answered on behalf of their household. This dataset constituted the empirical foundation for constructing and validating the Care4food Score, which was organised into four sub-domains: protein, fruit and vegetables, ultra-processed foods, and sustainability practices.

The baseline survey data were used for multiple purposes: (i) to examine empirical distributions of dietary behaviours and define realistic scoring thresholds (e.g. red-meat consumption cut-offs, legume intake frequency, and penalties for fruit–vegetable imbalance); (ii) to quantify discrepancies between perceived and behaviour-based dietary sustainability, providing early evidence of systematic overestimation consistent with Dunning–Kruger–type dynamics; and (iii) to conduct sensitivity analyses ensuring that the scoring system was interpretable and robust prior to its implementation in the digital survey environment.

The Care4Food score comprises four sub-scores. All sub-scores are based on harmonised frequency-response categories: never; occasionally during the month; once a week; a few times a week; every day; and several times a day, ensuring consistency across items.

Core survey items used to construct the sub-scores included: (1) self-reported dietary intake by food group, assessed through the question: “How often do you consume each of the following foods?”; and (2) food-related sustainability practices, including frequency of eating outside the home, frequency of purchasing fresh foods (fruit and vegetables), primary mode of transport for food shopping, and food-waste behaviours.

Table 1 summarises how each sub-score maps specific survey variables to rewarded or penalised behaviours, together with their alignment to international and national frameworks. This structure captures both nutritional adequacy and broader environmental and socio-cultural dimensions of dietary sustainability, while enabling direct comparison between behaviour-based scores and participants’ self-perceived sustainability. Penalties were applied to specific dietary behaviours that diverge from evidence-based recommendations on sustainable healthy diets (FAO & WHO, 2019; Willett et al., 2019; Polzin et al., 2023; Neta et al., 2023). This was the case for fruit and vegetable intake when one group was consumed frequently and the other rarely, reflecting dietary imbalance, and for sustainability practices such as systematic private-car use and frequent food waste, which are associated with disproportionate environmental impacts. This approach ensures that the scoring system not only rewards sustainable habits but also captures the negative effects of unsustainable ones, thereby supporting a more accurate and policy-relevant sustainability assessment.

Overall, the animal and vegetable protein balance sub-score reflects the balance between plant- and animal-based protein sources. Higher scores correspond to dietary patterns characterised by limited red-meat consumption, regular intake of plant proteins (e.g. legumes and cereals), and moderate consumption of animal products. Lower scores reflect frequent red-meat consumption and imbalanced reliance on a narrow set of protein sources.

The fruit and vegetables intake sub-score captures both the regularity and balance of fruit and vegetable consumption. High scores are assigned when both fruit and vegetables are consumed daily or several times per day.

Ultra-processed food (UPF) consumption was assessed using predefined survey items referring to commonly and culturally recognized UPFs categories in the Portuguese context, including sweet and salty snacks (*chips, salgados*), processed meats (*enchidos*), and ready-to-eat meals. The use of these culturally familiar categories, which are widely understood by respondents, corresponds conceptually to ultra-processed foods as defined in Group 4 of the NOVA food classification system and enhances interpretability and response reliability. The UPF intake sub-score is inversely related to the frequency of consumption of sweet and savoury snacks and ultra-processed foods. Never or very rare consumption yields high scores, whereas daily or several-times-daily consumption results in low scores

The sustainability practices sub-score captures food-related behaviours beyond dietary intake. These include the frequency of takeaway meals and eating out (with home cooking rewarded), food-purchasing locations (local markets and direct purchasing from producers rewarded), frequency of food waste, frequency of purchasing fresh fruit and vegetables, and transport modes used for food shopping (walking and cycling rewarded). Additional practices include preferences for buying nearby products and integrating food shopping into daily routines.

To ensure feasibility and reproducibility, survey responses were converted into scores ranging from 1 (least sustainable) to 5 (most sustainable) using predefined frequency-based thresholds. The scoring logic accounts for both positive behaviours (e.g. fruit and vegetable intake) and negative behaviours (e.g. ultra-processed food consumption), with additional adjustments applied for imbalanced dietary

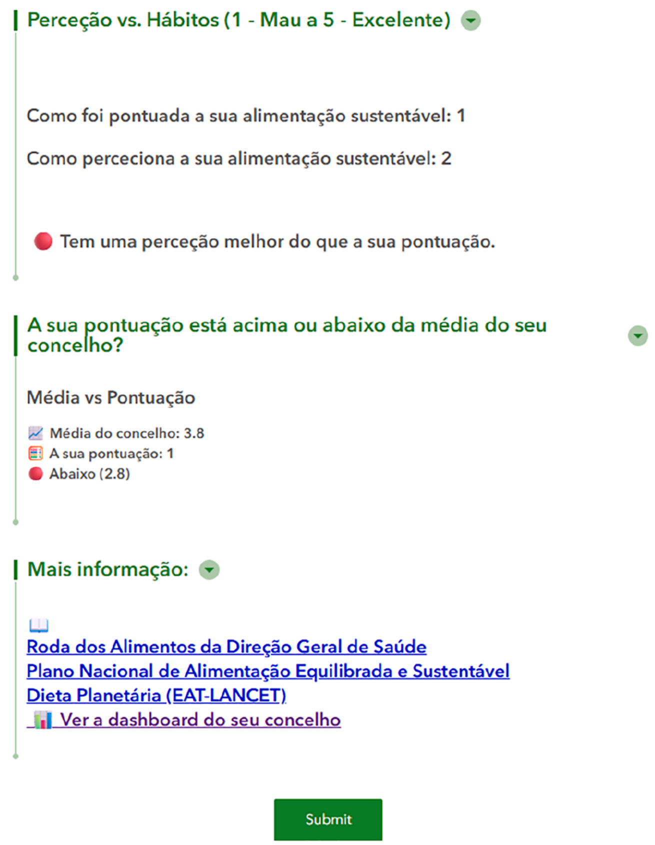

Fig. 2. Automated personalised feedback provided to survey participants. Caption: Example of the automated feedback interface displayed to participants upon completion of the Care4Food survey. The interface presents the behaviour-based Care4Food Score (1 = low, 5 = high), the participant’s self-perceived diet sustainability score, an interpretation of the perception–behaviour gap, and a comparison with the municipal average. The interface also provides links to international and national dietary guidelines, supporting reflection and information without prescriptive messaging.

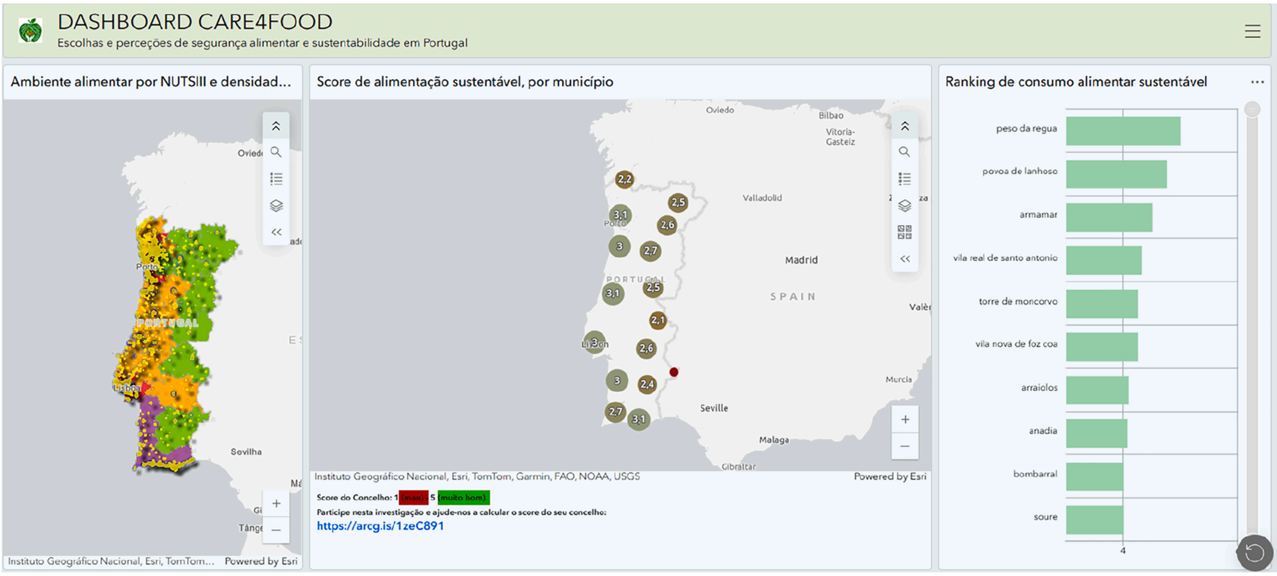

Fig. 3. Care4Food dashboard for exploratory monitoring of sustainable diets in Portugal.
Caption: the Care4food dashboard integrates citizen-generated dietary data with contextual food-environment indicators. The left panel displays food-environment characteristics at the NUTS III level, the central map shows the mean behaviour-based Food Sustainable Score aggregated by municipality, and the right panel presents a ranking of municipalities based on mean scores. Indicators are updated dynamically as new submissions are received and are intended for descriptive visualisation and exploratory spatial analysis supporting prioritisation and hypothesis generation rather than population-representative inference.

patterns. A complete description of the scoring algorithm, including conversion tables and decision rules, is provided in Supplementary Material S1.

The Care4Food score (*score_geral*) is the arithmetic mean of the four sub-scores:

$$\text{Care4Food Score } (score\_geral) = \frac{\text{scoreprotein} + \text{scorefruit} + \text{scoreprocessed} + \text{scoresustentabilidade}}{4} \tag{1}$$

The composite Care4Food score was calculated as the unweighted mean of the four sub-scores. Equal weighting was adopted as a reasonable and transparent approach, reflecting the conceptual importance of each domain within the FAO/WHO framework for sustainable healthy diets. Given the absence of a universally accepted weighting scheme, and that sensitivity analyses indicated that alternative weighting schemes did not materially affect results, this approach prioritises interpretability across dimensions.

To capture participants' perceptions of their own dietary sustainability, respondents were asked: *"Do you consider that you follow a healthy and sustainable diet?"* Responses were recorded on a five-point Likert scale ranging from 1 (*not at all healthy and sustainable*) to 5 (*very healthy and sustainable*). This variable is referred to as perceived diet sustainability ($\text{dieta}_{\text{percecionada}}$).

Self-assessed measures of this type are subject to social desirability bias and overconfidence, whereby individuals may overestimate the sustainability of their diets (de Ridder et al., 2012). To assess miscalibration, a perception–behaviour gap was calculated as the difference between self-perceived diet sustainability and the behaviour-based Care4Food score as:

$$\Delta = dieta_{percecionada} - score\_geral \tag{2}$$

Positive values ($\Delta > 0$) indicate overestimation, negative values indicate underestimation, and values close to zero reflect accurate self-assessment. This formulation preserves the conceptual meaning of the perception scale while enabling systematic comparison with the behaviour-based indicators.

Insights from this calibration phase informed the final scoring rules implemented in the VGI-based digital survey and were subsequently used to generate real-time feedback and support spatial aggregation in the online deployment.

Table 2
Distribution of mean sub-scores by quartile of the Care4Food score (mainland Portugal, n = 1813).

Care4Food quartile	N	%	Cum. %	Care4Food score mean	Care4Food score Std. Dev.	Protein	Fruit & veg	Ultra processed	Sustainability practices
Q1 (lowest)	389	21.5	21.5	1.96	0.3	1.64	2.34	2.47	1.41
Q2	561	30.9	52.4	2.75	0.21	2.46	3.27	3.09	2.19
Q3	383	21.1	73.5	3.37	0.12	3.18	3.76	3.44	3.08
Q4 (highest)	480	26.5	100	4.13	0.37	4.14	4.31	3.94	4.12
Total	**1813**	**100**		**3.08**	**0.82**	**2.88**	**3.45**	**3.26**	**2.72**

2.4. Digital implementation and municipal aggregation

Following score construction and calibration using the baseline survey (n = 1813), the questionnaire and scoring logic were implemented as an XLSForm and deployed via a web-based geospatial survey tool (ArcGIS Survey123) to enable open, voluntary participation. The baseline dataset was used to initialise the online system with reference values and to test the end-to-end workflow (see Supplementary Material S1 – Appendix A).

A pilot phase was conducted to assess:(i) comprehension of survey items in the mobile interface; (ii) correct execution of XLSForm calculations for the four sub-scores and the composite Score; (iii) municipality assignment procedures; (iv) reliable storage of submissions in a secure geospatial database; and (v) clarity and usability of the automated feedback presented to respondents. Following minor adjustments, deterministic scoring rules were embedded in the digital questionnaire, enabling automatic computation of sub-scores and composite indicators (range 1–5) at the time of submission.

To preserve anonymity while enabling territorial aggregation, each submission was linked to a municipality, and geographic information was retained only at municipality-centroid resolution, with no precise coordinates stored. Responses were summarised to generate municipal-level indicators, including mean sub-scores, the municipal mean Care4Food score, distributions of perception-behaviour gaps (Δ), and participation counts.

A central component of the digital implementation is the provision of immediate personalised feedback upon survey completion. Each participant receives: (i) their behaviour-based Care4Food Score; (ii) their perceived diet sustainability score; (iii) a qualitative interpretation of alignment or mismatch between perception and behaviour; (iv) a comparison with the average score of their municipality; and (v) links to authoritative dietary guidance, including the Portuguese Food Wheel, the National Sustainable Diet Plan, and EAT–Lancet recommendations (Fig. 2).

Survey data are updated dynamically as new submissions are received. Aggregated indicators are visualised through an interactive public dashboard that integrates citizen-reported dietary data with contextual food-environment layers, namely: (i) food-environment indicators (e.g. food retail density and fast-food presence); (ii) a municipality-level map of mean Care4Food scores; and (iii) a ranking of municipalities based on mean scores (Fig. 3).

Together, the survey and the dashboard constitute a citizen-driven monitoring prototype that links individual dietary behaviours and perceptions to municipal-level descriptive outputs. The dashboard is not presented as a technological product, but as a governance interface that translates individual-level behavioural data into territorially aggregated signals interpretable within municipal policy competences.

Because the baseline survey was quota-sampled at the NUTS II level rather than at the municipal level, all inferential statistical analyses are conducted at the individual level. Municipal-level indicators are used exclusively for descriptive visualisation and demonstration of the monitoring prototype, benchmarking, and policy learning, not for population-representative inference. Descriptive and inferential statistical analyses were conducted using IBM SPSS Statistics, version 29, and included descriptive statistics, cross-tabulations, chi-square tests, and correlation analyses, with statistical significance assessed at the 5% level unless otherwise stated. This research was approved by the Ethics Committee of the University of Lisbon in accordance with national and European guidelines, and written informed consent was obtained from all participants.

3. Results

3.1. Distribution of the Care4food score and sub-scores

This analysis included 1813 respondents, each representing a household. The mean age of participants was 50 years, consistent with the minimum eligibility age of 18 years. The sample comprised 53% women and 47% men. Most household heads were employed (64%), while 26% were retired and 6% unemployed. Unemployment was highest in the Península de Setúbal (12%), followed by the Metropolitan area of Porto (8%). Regarding self-reported dietary patterns, 66% of participants reported diets rich in fat, dairy, and protein. Mediterranean and flexitarian diets were more prevalent in southern regions, particularly in the Algarve (39%), whereas special diets (vegetarian, vegan, or other specific dietary patterns) accounted for 3% of the sample and were more frequently reported in metropolitan areas.

The Care4Food Score (range 1–5) showed a moderate distribution across participants, with a mean of 3.08 (SD = 0.82) and a median of 3.0, indicating substantial heterogeneity in adherence to sustainable dietary

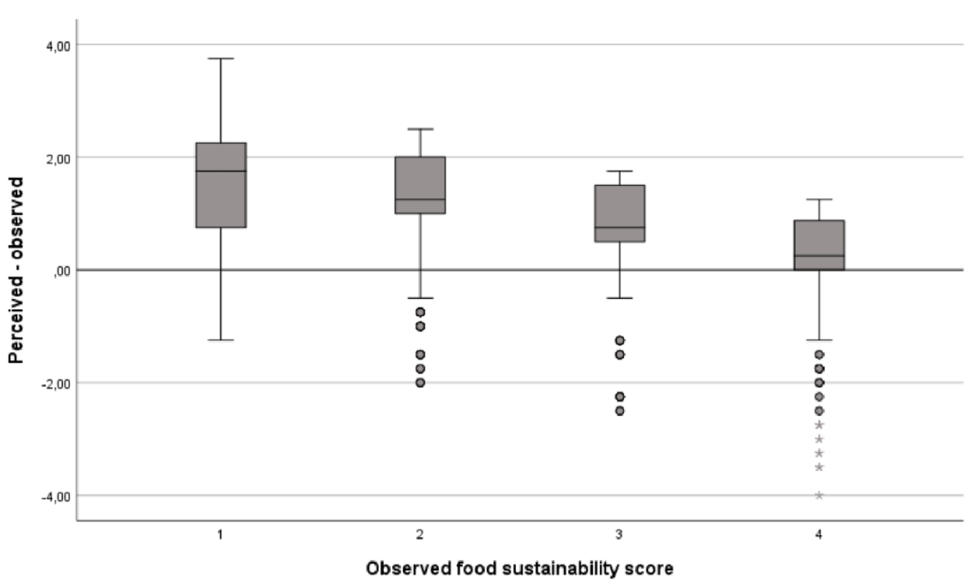

Fig. 4. Perception–behaviour miscalibration in dietary sustainability across score quartiles.
The figure shows the difference between perceived and behaviour-based dietary sustainability scores (y-axis: perceived − behaviour-based score) across quartiles of the behaviour-based Care4Food Score (x-axis). Positive values indicate overestimation of dietary sustainability. Participants in the lowest quartile exhibit substantial overestimation, whereas those in the highest quartile display more accurate or slightly conservative self-assessment.

practices. In addition, median-based analyses yielded results consistent with mean-based analyses, indicating that findings are not driven by extreme values. Participants in the lowest quartile (Q1) represented 21.5% of the sample, while those in the highest quartile (Q4) accounted for 26.5% (n = 480) (Table 2).

As shown in Table 2, mean sub-scores increased systematically across quartiles of the Care4Food score, enabling identification of the dietary components that most strongly differentiate low- and high-sustainability profiles. Among the sub-dimensions, fruit and vegetable intake showed the highest overall adherence (mean = 3.45), followed by lower consumption of ultra-processed foods (mean = 3.26). In contrast, protein balance (mean = 2.88) and sustainability-oriented practices that include local food purchasing, active transport to food outlets, and food-waste avoidance (mean = 2.72), displayed the lowest adherence and contributed most strongly to lower overall sustainability scores. The Care4Food score was moderately correlated with all sub-scores, with the strongest association observed for sustainability practices (Pearson r = 0.66).

Adequate fruit and vegetable intake (scores 4–5) was observed in 56.7% of participants, whereas 22.4% exhibited low intake levels (scores 1–2). For ultra-processed foods, 32.4% of participants recorded low sub-scores, reflecting frequent consumption, while 46.8% achieved favourable scores indicative of limited intake.

Protein balance emerged as a critical weakness: 49.6% of participants scored in the lowest categories (scores 1–2), reflecting dietary patterns characterised by frequent red-meat consumption, whereas only 35.7% achieved balanced protein scores (scores 4–5). Adoption of sustainability-oriented practices was similarly uneven, with 35.0% of participants in the lowest category and 36.0% in the highest (scores 4–5).

Care4Food score distributions differed significantly by gender ($\chi^2(3) = 47.52$, $p < 0.001$), with women more frequently represented in the highest sustainability quartile. Distributions also differed significantly by health status ($\chi^2(3) = 18.76$, $p < 0.001$), as participants reporting health conditions were more often classified in the highest quartile (32.3%) compared with those reporting no health conditions (24.7%). Dietary profiles among participants in the highest quartile were broadly consistent with previously described sustainable dietary patterns, although differences by socio-economic characteristics persisted.

Taken together, these findings indicate that lower overall Care4Food scores were primarily driven by imbalanced protein intake, frequent consumption of ultra-processed foods, and limited adoption of sustainability-oriented practices, while fruit and vegetable intake played a comparatively smaller role in differentiating sustainability profiles.

3.2. Perceived score versus behaviour-based Care4Food score

Participants generally perceived their diets as healthy and sustainable. On average, participants rated their diet sustainability relatively high (mean perceived = 4.0, SD = 1.1), while the behaviour-based Care4Food Score was lower (mean = 3.08, SD = 0.82). As a result, the perception–behaviour gap ($\Delta = dieta_{percepcionada} - score\_geral$) was positive on average, indicating systematic overestimation of dietary sustainability. Perceived sustainability showed only a weak correlation with the behaviour-based score (Pearson r = 0.29), suggesting limited calibration between self-assessment and reported dietary behaviours.

Table 3
- Behaviour-based and perceived dietary sustainability scores by gender in the online deployment.

	Gap	Behavior-based Care4food score	Perceived score	Count
Men	1.62	2.92	4.53	15
Women	0.99	3.05	4.04	25
Total	**1.23**	**3.00**	**4.23**	**40**

A strong negative association was observed between the behaviour-based Care4Food Score and miscalibration (Spearman's ρ = −0.49, 95% CI −0.53 to −0.45; p < 0.001). Participants with lower sustainability scores tended to substantially overestimate the sustainability of their diets, whereas those with higher scores demonstrated more accurate or slightly conservative self-assessments. This pattern is consistent with a Dunning–Kruger-type effect in dietary behaviour.

Stratified analyses indicated that this miscalibration pattern was consistently present across population subgroups. The negative association between the behaviour-based score and miscalibration was observed among both men (ρ = −0.39, p < 0.001) and women (ρ = −0.59, p < 0.001), with a stronger association among women. Similar patterns were identified among participants with health conditions (ρ = −0.56, p < 0.001) and without health conditions (ρ = −0.47, p < 0.001), as well as across all education levels. Effect sizes ranged from moderate to strong, with no consistent attenuation at higher levels of educational attainment.

Overall, participants in the lowest Care4Food score quartile substantially overestimated the sustainability of their diets, whereas those in the highest quartile tended to assess their diet sustainability more accurately or to slightly underestimate it (Fig. 4). These findings reveal a pronounced perception–practice gap in dietary sustainability and highlight the value of the Care4Food Score for identifying misperceptions that may hinder effective dietary change and targeted public health interventions.

3.3. Validation and robustness of the Care4Food score

The validity and robustness of the Care4Food score were assessed through sensitivity analyses, internal consistency, inter-item correlations, convergent validity, and construct validity. These complementary approaches provide evidence of both the internal structure and external relevance of the index as a multidimensional composite indicator of dietary sustainability.

Sensitivity analyses were conducted by recalculating the composite Care4Food score using alternative weighting schemes that assigned different relative importance to the four sub-scores. A sustainability-focused direction (weights: protein = 0.20; fruit and vegetables = 0.20; ultra-processed foods = 0.20; sustainability practices = 0.40) yielded a very high correlation with the original score (r = 0.963), while a nutrition-focused direction (weights: protein = 0.30; fruit and vegetables = 0.30; ultra-processed foods = 0.30; sustainability practices = 0.10) also showed a similarly strong correlation (r = 0.955). These results indicate that the Care4Food score is highly robust to alternative weighting assumptions, with minimal impact on the ranking of individuals or the overall distribution of scores.

Internal consistency across the four sub-scores was low (Cronbach's α = 0.29), reflecting the multidimensional structure of the index. As the Care4Food score integrates conceptually distinct domains of dietary sustainability rather than measuring a single latent construct, lower internal consistency is expected and does not indicate poor performance of the index. Additionally, the correlations between sub-scores were low (r ≈ 0.02–0.21), indicating that the components capture distinct and complementary dimensions of dietary sustainability rather than reflecting a single underlying construct.

Convergent validity, defined as the extent to which specific sub-dimensions correlate with the composite score, was supported by moderate to strong correlations: r = 0.51 for ultra-processed foods, r = 0.52 for fruit and vegetables, r = 0.58 for protein balance, and r = 0.65 for sustainability practices.

Construct validity was supported by statistically significant differences in Care4Food scores across socio-demographic groups. Women exhibited higher scores than men (mean = 3.20 vs. 2.94; t = −6.99, p < 0.01), consistent with established patterns in dietary behaviour (Feraco et al., 2024).

Taken together, these results support the validity and robustness of

Fig. 5. - Behaviour-based Care4Food Score by municipality.
Caption: Map showing the mean Care4food Score aggregated at the municipal level from voluntary online submissions. Scores range from 1 (low sustainability) to 5 (high sustainability) and are calculated as the mean of individual responses within each municipality. Values are presented for descriptive and exploratory purposes only, as participation levels vary across municipalities (visual representation with a minimum threshold of 5 responses).

the Care4Food score as a multidimensional, behaviour-based indicator that captures meaningful variation in dietary sustainability and is suitable for monitoring in a policy-relevant context.

3.4. Scaling individual dietary data to municipal-level indicators

Following deployment of the validated scoring system in the online survey, a total of 40 voluntary submissions were collected during the initial online phase by December 2025.

Submissions were spatially concentrated in the Metropolitan area of Lisbon (70% of responses), with lower participation in other NUTS II regions. Consequently, municipal and regional indicators derived from the online phase are presented only as descriptive outputs to assess the functional performance of the monitoring prototype, with emphasis on automated score computation, personalised feedback delivery, and live spatial aggregation, rather than on individual- or population-level inference. As participation increases, the system is designed to support progressively finer-grained spatial aggregation and analysis.

All submissions successfully generated the four sub-scores and the composite behaviour-based Care4Food Score at the time of survey completion. Participants received immediate personalised feedback comparing their behaviour-based score with their self-perceived sustainability and with the mean score of their municipality, confirming the correct operation of the XLSForm logic, scoring procedures, and dashboard integration.

Across online submissions, the mean behaviour-based Care4Food Score was 3.00, while the mean perceived sustainability score was higher (4.23), resulting in an average perception–behaviour gap of +1.23. This pattern closely mirrors results observed in the baseline survey, indicating that the citizen-science deployment reproduces the same systematic tendency toward overestimation of dietary sustainability identified under controlled sampling conditions (Table 3).

Gender-stratified results showed a larger perception–behaviour gap among men (mean gap = 1.62; behaviour-based score = 2.92; perceived score = 4.53) than among women (mean gap = 0.99; behaviour-based score = 3.05; perceived score = 4.04). Although based on a small sample, this directional consistency with baseline findings suggests that the system captures stable behavioural signals, rather than random variation.

Preliminary visualisation of municipal indicators suggests spatial heterogeneity in Care4Food Score, with lower values observed in some rural and inland areas compared with coastal municipalities. These patterns are exploratory and must be interpreted with caution; however, they demonstrate the system's capacity to generate spatially explicit outputs that can inform subsequent analytical and policy-oriented work, including targeting and prioritising municipal interventions (Fig. 5).

Overall, the online deployment demonstrates that individual-level behavioural and perceptual data can be reliably transformed into municipal-level indicators within a real-time geospatial framework, reproducing key behavioural patterns observed in the baseline survey.

4. Discussion

4.1. What the Care4Food score captures relative to established indices

Research on sustainable diets consistently emphasises that no single dimension is sufficient to characterise dietary sustainability and that robust indices must integrate nutritional, environmental, economic, and behavioural components (Rei et al., 2025; Rockström et al., 2025). In this context, the convergence observed between the Care4Food score and established composite indices suggests that the proposed indicator captures a comparable underlying construct of dietary sustainability.

When rescaled to a 0–20 metric, the mean Care4Food score observed in the baseline survey (≈ 12.3) closely matches reported values for the Sustainable Diet Index (SDI) in the NutriNet-Santé cohort (≈ 12.1) and its U.S. adaptation (≈ 13.1) (Jung et al., 2024; Seconda et al., 2020). This similarity is noteworthy given the substantial methodological differences between the tools. Whereas the SDI relies on detailed dietary intake data, life-cycle-based environmental footprint modelling, and economic cost estimation, the Care4Food score is derived from simplified food-frequency questions and self-reported food-related practices. Similarity in central tendency does not imply equivalence between indices. Rather, it indicates that both are grounded in shared normative frameworks, particularly the FAO/WHO Guiding Principles for Sustainable Healthy Diets and the EAT–Lancet Planetary Health Diet, which integrate nutritional, environmental, economic, and sociocultural dimensions.

At the population level, these frameworks converge on common expectations: high consumption of plant-based foods, limited red and processed meat intake, moderate dairy consumption, reduced ultra-processed foods, and supportive food-related practices. In European contexts, these expectations typically translate into partial rather than full adherence, a pattern consistent with the distribution observed in this study. Analysis of sub-dimensions further supports this interpretation. In both the SDI and the Care4Food score, environment-related components show the strongest association with overall sustainability. This suggests that food-related practices play a particularly important role in differentiating more sustainable from less sustainable diets. In practice, many everyday food-related behaviours simultaneously reflect multiple dimensions of sustainability. For example, reduced consumption of ultra-processed foods relates not only to nutritional quality but also to environmental impact and food-system characteristics, while practices such as local food purchasing or food-waste reduction integrate environmental and socio-cultural aspects.

Positioning this analysis within the landscape of existing indices further clarifies the specific contribution of the Care4Food approach. Established indices such as the SDI and the Planetary Health Diet Index (PHDI) rely on detailed dietary intake data (e.g. 24-hour recalls or food diaries) combined with environmental and nutritional modelling. While these approaches provide high-resolution assessments, they are resource-intensive and generally limited to research settings or periodic surveys.

By contrast, the Care4Food score adopts a simplified, behaviour-based approach grounded in food-frequency questions and observable practices. This involves a trade-off between analytical precision and operational feasibility. Although it does not capture exact quantities or life-cycle impacts, it enables low-cost, scalable, and repeatable data collection suited to routine monitoring at municipal and regional levels. Care4Food should therefore be understood as a complementary tool that translates established sustainability principles into practical, policy-relevant indicators.

Importantly, the inclusion of a perception-based measure allows the identification of perception–behaviour gaps, a dimension largely absent from existing indices but critical for understanding behavioural barriers and informing policy interventions. The integration of behavioural, perceptual, and spatial dimensions thus represents a key added value of the proposed approach.

4.2. Perception-practice mismatch as a central behavioural finding

A central contribution of this study is the explicit measurement of the gap between perceived and behaviour-based dietary sustainability. Across the baseline survey, participants consistently rated their diets as more sustainable than indicated by their reported behaviours, with only a weak correlation between self-perception and objective scoring. This miscalibration followed a clear gradient: individuals with the lowest sustainability scores exhibited the greatest overestimation, whereas those with higher scores assessed their diets more accurately or conservatively. Importantly, the online citizen-science deployment reproduced the same perception-behaviour gap, including stronger overestimation among men, mirroring baseline results. This pattern is consistent with behavioural-science evidence on self-assessment biases, including Dunning-Kruger-type effects previously documented in diet quality, physical activity, and health literacy (Menozzi et al., 2023; Thomson et al., 2023; Worsley et al., 2014). This consistency suggests that the observed mismatch is not a sampling artefact but a stable behavioural pattern with implications for the design, targeting, and evaluation of diet-related policies across governance levels. Thus, from a public policy perspective, these results are particularly relevant for interventions implemented at local and sectoral levels, such as nutrition education programs, food-literacy initiatives, and sustainable consumption campaigns (e.g. waste mitigation) which often rely on information provision and voluntary behaviour change. If individuals substantially overestimate the sustainability of their diets, motivation to adjust behaviour may be limited, reducing the effectiveness of awareness-based strategies.

4.3. From individual data to spatially explicit and integrated policy support

This study demonstrates how citizen-generated data can complement traditional food-system monitoring by introducing a spatially explicit, behaviour-based layer that connects everyday dietary practices to territorial governance. Beyond producing a simplified sustainable diet score, the Care4Food approach addresses a structural challenge widely identified in urban and regional planning studies: policy fragmentation across sectors and scales.

In the Portuguese context, responsibilities related to food and diets are distributed across spatial planning, health, education, agriculture, and environmental policy domains. Municipalities manage land-use planning and local accessibility, while sectoral frameworks require local authorities to implement sustainable public food procurement schemes, such as school meal programs, under European guidance. At the same time, regional and local health authorities promote dietary guidance and food literacy through the National Program for the Promotion of Healthy Eating, and agricultural policies encourage short supply chains, food-waste reduction, and agroecological transitions within broader EU strategies. Despite overlapping objectives across governance scales, these domains rarely share monitoring instruments capable of supporting coordinated place-based action (Oliveira, 2022; Arcuri et al., 2022; Abrantes et al., 2025).

By aggregating individual dietary behaviours and food-related practices at municipal and supra-municipal levels, Care4Food provides a shared territorial evidence base that can be interpreted simultaneously within health, spatial planning, agrifood, and sustainability policy sectors. Indicators such as protein balance, ultra-processed food consumption, local purchasing practices, food waste, and transport to food outlets intersect directly with competences and interests exercised at city, metropolitan, or metropolitan/regional levels. As a result, the system supports horizontal coordination across sectors and vertical alignment across governance scales.

At the European level, this approach responds to calls for integrated food-system governance articulated in the EU Farm to Fork Strategy and related initiatives, which emphasise linkages between health, environment, agriculture, and territorial development while offering limited operational guidance for sub-national diet-related monitoring. By operationalising framework-aligned indicators at territorial scales, Care4Food illustrates how EU-level objectives can be translated into spatially grounded monitoring tools that support benchmarking, cross-territorial comparison, and multi-level policy coordination (Andrews & Sanderson Bellamy, 2023).

Although current territorial outputs remain descriptive due to participation levels, early dashboard results replicate baseline patterns and reveal spatial heterogeneity across coastal, metropolitan, and more rural areas. These exploratory patterns demonstrate the added value of spatialised citizen-generated data. Rather than producing static reports, the system enables continuous updating of diet-related behavioural signals, supporting prioritisation, hypothesis generation, and policy learning without claiming causal inference.

Thus, from an operational perspective, the Care4Food system supports routine, low-cost monitoring rather than one-off assessments. For policy use, repeated data collection at regular intervals would allow municipalities and sectors to track trends over time, identify priority domains (e.g. protein imbalance or high ultra-processed food consumption), monitor perception–behaviour gaps, support targeted interventions, and benchmark progress across territories.

Importantly, its value lies not in precise dietary measurement but in its ability to provide continuous, interpretable behavioural signals aligned with policy competences. Minimum sample thresholds are necessary to ensure meaningful interpretation of municipal-level indicators; in contexts with lower participation, results should be interpreted as exploratory signals rather than representative statistics.

As participation expands, the approach enables testing of policy-relevant hypotheses, for example, whether territories with stronger food environments, local markets, or food-literacy initiatives display smaller perception–practice gaps. The system is explicitly designed to scale, replicate, and integrate systemic data on food environments, agrifood systems, socio-economic conditions, and local policy interventions.

More broadly, information remains central to food-policy design. However, information alone is insufficient to drive sustainable dietary transitions. Monitoring tools that explicitly capture perception–practice gaps and territorial variation can support more realistic and targeted interventions, particularly given that sustainability-related dimensions of diets remain less visible and less systemically understood than nutritional health among citizens and stakeholders (Aiking, 2004; García-González et al., 2022).

4.4. Implications for municipal sustainable food-systems planning and governance

Cities, metropolitan areas, and regions play an increasingly important role in advancing sustainable diets through public food procurement, school and institutional catering, health promotion, local food initiatives, waste programs, and spatial-planning decisions that shape food environments (Pertoldi et al., 2022). Yet sub-national authorities typically rely on macro-level indicators or infrequent surveys that provide limited insight into everyday dietary behaviours and citizens' perceptions.

The Care4Food approach addresses this governance gap by translating individual food-related practices into territorially aggregated indicators aligned with policy competences. By digitalising citizen-reported dietary behaviours, the system makes visible dimensions of sustainable diets that are largely absent from official statistics, including protein sourcing patterns, reliance on ultra-processed foods, food-waste practices, local purchasing behaviours, and mobility-related aspects of food access.

These indicators are directly interpretable within urban and regional policy domains, including nutrition education, food-access planning, support for short supply chains, sustainable mobility strategies, and procurement improvements. Unlike highly technical composite indices, the Care4Food score is based on transparent, behaviourally meaningful components that can be communicated to citizens, practitioners, and decision-makers.

Importantly, the systematic perception-behaviour gap identified in both the baseline survey and the online deployment has direct implications for sub-national policy design. Many urban and regional interventions, such as nutrition education programmes, food-literacy initiatives, or sustainable consumption campaigns, implicitly assume that citizens accurately perceive the sustainability of their diets. The observed overestimation, particularly among lower-scoring individuals, suggests that awareness-based strategies alone may have limited impact if misperceptions are not addressed. By combining behaviour-based indicators with self-perceived sustainability measures, the system enables local and regional authorities to identify where information deficits, overconfidence, or uneven knowledge may undermine policy effectiveness. Immediate personalised feedback further strengthens the educational function of the tool, supporting citizen reflection while simultaneously generating aggregated territorial data.

The low-burden digital architecture supports routine monitoring, benchmarking across cities and regions, and longitudinal tracking of change over time, without the financial and logistical demands of traditional dietary surveys. While current territorial outputs are descriptive, the system outlines a pathway toward more institutionalised but inclusive participatory monitoring frameworks.

Overall, the Care4Food score appears realistically calibrated, capturing meaningful signals of dietary sustainability despite relying on lower-cost, citizen-generated data. This supports its use as a pragmatic proxy for sustainable dietary behaviour in contexts where resource-intensive dietary assessment is not feasible. By aligning individual-level behavioural evidence with public policy's needs, the approach contributes to more integrated, participatory, and spatially sensitive models of sustainable food governance across cities, metropolitan areas, and regions.

4.5. Limitations and future directions

The Care4Food score is based on self-reported behaviours and is therefore subject to bias. Respondents may over-report behaviours perceived as sustainable, such as fruit and vegetable consumption or reduced intake of ultra-processed foods, while under-reporting less desirable practices. Additionally, respondents may focus primarily on health-related aspects of diet while overlooking broader sustainability dimensions, such as carbon footprints, seasonality, or red-meat consumption. These factors may introduce bias into behaviour-based scores and contribute to the perception–behaviour gap observed in this study.

The use of a single self-assessment question combining "healthy" and "sustainable" diet concepts may also introduce ambiguity, as respondents may not perceive these dimensions as equivalent. Some individuals may consider their diet healthy but not environmentally sustainable, or may be uncertain about sustainability concepts. However, this limitation is partially mitigated by the structure of the survey, which includes multiple behaviour-based questions on food consumption patterns and food-related practices. These items act as internal control variables, providing an independent basis for constructing the Care4Food score and reducing the influence of biases or inconsistencies in self-perceived responses. Empirical results show a significant association between reported meal/diet types and perceived diet sustainability ($p = 0.01$), supporting the internal coherence of the instrument. As a result, discrepancies between perceived and behaviour-based scores can be interpreted not only as measurement noise but also as meaningful indicators of perception–behaviour gaps. However, future research could refine this component by separating health and sustainability perceptions into distinct questions.

From a methodological perspective, although the use of frequency-based categories and familiar food-group questions reduces cognitive burden and improves response consistency, future implementations could incorporate calibration using subsamples with more detailed dietary assessments or integrate objective proxies (e.g. purchase data or food-environment indicators) where available. In addition, establishing minimum sample thresholds for public reporting, exploring weighting or smoothing procedures, and developing strategies to sustain long-term participant engagement will be essential to ensure that municipal-level indicators evolve beyond descriptive outputs and gain analytical robustness.

As the analysis is based on data from mainland Portugal, the findings may not be directly generalisable to other countries with different dietary cultures, food environments, and socio-economic contexts. While the use of the mean supports interpretability and comparability, future applications could also consider median-based reporting in contexts with highly skewed distributions.

Finally, to support correct interpretation, future implementations should be accompanied by simple user guidelines explaining the meaning of score ranges, the contribution of sub-dimensions, and the limitations of interpreting the composite score in isolation.

Overall, citizen-generated data provide valuable insights that traditional sources may not capture and offer a useful perspective for strengthening food-sustainability literacy, particularly regarding environmental and systemic aspects of diets. However, as with other forms of crowdsourced data, participation levels may fluctuate over time. Increased coverage can improve completeness while simultaneously introducing variability and precision challenges (Lukyanenko, 2019).

Addressing these trade-offs will be critical for the long-term viability of participatory diet-related monitoring within broader food-system governance.

5. Conclusions

This study achieved its three main objectives. First, a behaviour-based sustainable diet score aligned with international frameworks was successfully constructed and calibrated using a large-scale survey. Second, the analysis identified a consistent perception–behaviour gap, with participants systematically overestimating the sustainability of their diets. Third, the study demonstrated the feasibility of translating citizen-generated data into spatially explicit, municipal-level indicators through a digital geospatial monitoring system. Together, these results confirm the potential of the Care4Food approach as a practical tool for policy-relevant monitoring of sustainable diets.

As a behaviour-based, citizen-generated scoring system, Care4Food can meaningfully capture sustainable dietary patterns when anchored in established international frameworks. Compared with more resource-intensive sustainable diet indices, it offers a lower-burden and more scalable alternative for territorial monitoring. It translates FAO/WHO, EAT–Lancet, and national dietary guidance into a food-group-based index that balances nutritional quality, environmental relevance, and everyday food practices. Its convergence with more data-intensive indices, such as the Sustainable Diet Index, supports the construct validity and realistic calibration of the proposed approach.

By embedding individual-level behavioural data within a spatially explicit dashboard and enabling aggregation at municipal and supra-municipal levels, Care4Food connects dietary practices to territorial governance. It offers a pragmatic and scalable monitoring instrument for cities, metropolitan areas, and regions seeking to align food, health, and sustainability objectives, operationalising behavioural sustainability monitoring within a smart-governance framework and translating citizen-generated data into territorially interpretable signals for policy coordination.

Overall, the results show how simplified, transparent, and spatially grounded scoring systems can complement existing food-system monitoring and support more integrated, policy-relevant, and territorially sensitive approaches to dietary sustainability assessment and decision-making.

Future work will focus on expanding participation to improve territorial coverage, validating the score across different cultural and geographic contexts, and integrating additional data sources to enhance robustness and user interpretability.

Use of generative AI and AI-assisted technologies

Statement: During the preparation of this work, the author(s) used ChatGPT (OpenAI) to assist with language editing and the checking and correction of technical XLSForm code used for survey implementation. After using this tool, the author(s) reviewed and edited the content as needed, taking full responsibility for the content of the published article.

Funding

This research was supported by the Fundação para a Ciência e a Tecnologia (FCT) through multiannual funding for the period 2025–2029 (UID/295/2025).

Ethical approval

This study was approved by the Ethics Committee of the University of Lisbon in accordance with national and European guidelines. Written informed consent was obtained from all participants.

CRediT authorship contribution statement

Patrícia Abrantes: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Eduarda Marques da Costa:** Writing – original draft, Methodology, Investigation.

Declaration of competing interest

The authors declare no conflicts of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scsadv.2026.100069.

Data availability

Data will be made available on request.

References

Abrantes, P., Moreno, L., Fonseca, M. L., Türeli, B. B., Stergiadis, C., Szüdi, G., … Zorell, C. V. (2025). Examining inclusive governance of community gardens: Lessons from European cities. *City and Environment Interactions*, Article 100286.

Adewopo, J. B., Abayomi, A., Ogunmodede, T. A., Arbia, G., Genovese, G., & Amerise, I. L. (2025). *Crowdsourced food price data for near–real-time monitoring: Evidence from nigeria*. World Bank–FAO Collaborative Study.

Afshin, A., Sur, P. J., Fay, K. A., Cornaby, L., Ferrara, G., Salama, J. S., … Murray, C. J. (2019). Health effects of dietary risks in 195 countries, 1990–2017: A systematic analysis for the Global Burden of Disease Study 2017. *The lancet, 393*(10184), 1958–1972.

Aiking, H., & de Boer, J. (2004). Food sustainability: Diverging interpretations. *British Food Journal, 106*(5), 359–365. https://doi.org/10.1108/00070700410531618

Andrews, E., & Sanderson Bellamy, A. (2023). Putting food in the driver's seat: Aligning food-systems policy to advance sustainability, health, and security. *Frontiers in Sustainable Food Systems, 7*, Article 1187456. https://doi.org/10.3389/fsufs.2023.1187456

Arbia, G., Solano-Hermosilla, G., Nardelli, V., Micale, F., Genovese, G., & Adewopo, J. (2023). From mobile crowdsourcing to crowd-trusted food price data: Statistical preprocessing and post-sampling. *Scientific Data, 10*(1), Article 446. https://doi.org/10.1038/s41597-023-02278-9

Arcuri, S., Minotti, B., & Galli, F. (2022). Food policy integration in small cities: The case of intermunicipal governance in Lucca, Italy. *Journal of Rural Studies, 89*, 287–297.

Aylward, B. L., Milford, K. M., Storey, K. E., Nykiforuk, C. I., & Raine, K. D. (2022). Citizen science in monitoring food environments: A qualitative collective case study. *BMC Public Health, 22*(1), Article 665. https://doi.org/10.1186/s12889-022-13089-1

Cacau, L. T., de Carli, E., de Carvalho, A. M., Lotufo, P. A., Moreno, L. A., Bensenor, I. M., & Marchioni, D. M. (2021). Development and validation of the Planetary Health Diet Index. *Nutrients, 13*(5), Article 1698. https://doi.org/10.3390/nu13051698

Cirone, A., Moragues-Faus, A., Kneafsey, M., & Sonnino, R. (2023). Co-producing sustainability metrics for city-region food initiatives. *Sustainability, 15*(8), Article 6542. https://doi.org/10.3390/su15086542

Culliford, A., & Bradbury, J. (2020). A cross-sectional survey of public attitudes toward sustainable diets in the UK. *Appetite, 144*, Article 104444.

de Ridder, D., de Vet, E., Stok, M., Adriaanse, M., & de Wit, J. (2012). Obesity, overconfidence, and self-regulation failure. *Health Psychology Review, 6*(4), 415–436.

DGS (Direção Geral da Saúde). (2022). Pnpas. (2022–2030). *Programa nacional para a promoção da alimentação saudável*. Lisboa: DGS.

DGS (Direção Geral da Saúde). (2024). Programa Nacional de Promoção da Alimentação Saudável (PNPAS). *Dieta mediterrânica*. Alimentação Saudável. Lisboa: DGS. https://alimentacaosaudavel.dgs.pt/dieta-mediterranica/.

DGS (Direção-Geral da Saúde). (2016). *Roda dos alimentos mediterrânica*. Lisboa: DGS.

Dias, R. R., Sartori, R. B., & Schneider, A. T. (2024). Food Sustainability Index. *Smart food industry: The blockchain for sustainable engineering* (pp. 25–33). CRC Press.

Dickson-Spillmann, M., & Siegrist, M. (2011). Consumers' knowledge of healthy diets and its correlation with dietary behaviour. *Journal of Human Nutrition and Dietetics, 24*(1), 54–60.

Dijksterhuis, G., & Wallner, M. (2025). Eight easy pieces: Challenges for the science of food perception and behaviour. *Trends in Food Science & Technology, 159*, Article 104985.

European Commission. (2020). *Farm to fork strategy*. EC: Brussels.

Fanzo, J., Haddad, L., McLaren, R., Marshall, Q., Davis, C., Herforth, A., … Bellows, A. (2020). The Food Systems Dashboard is a new tool to inform better food policy. *Nature Food, 1*, 243–246. https://doi.org/10.1038/s43016-020-0077-y

FAO & WHO. (2019). *Sustainable healthy diets: Guiding principles*. Rome.

FAO, IFAD, UNICEF, WFP, & WHO. (2024). *The state of food security and nutrition in the world 2024*. Rome: FAO.

Feraco, A., Armani, A., Gorini, S., Camajani, E., Quattrini, C., Filardi, T., … Lombardo, M. (2024). Gender differences in dietary patterns and eating behaviours in individuals with obesity. *Nutrients, 16*(23), 4226.

The food systems dashboard. (2024). Global Alliance for Improved Nutrition, Columbia Climate School, & Cornell University College of Agriculture and Life Sciences (GAIN. https://doi.org/10.36072/db

García-González, Á., Achón, M., Krug, A. C., & Alonso-Aperte, E. (2022). Food sustainability knowledge and attitudes in the Spanish adult population: A cross-sectional study. *Nutrients, 14*(17), Article 3587. https://doi.org/10.3390/nu14173587

Harray, A. J., et al. (2022). Development of a healthy and sustainable diet index. *Public Health Nutrition, 25*(11), 3213–3225.

Harrington, R. A., et al. (2021). Harnessing crowdsourced food images for dietary assessment. *JMIR Public Health and Surveillance, 7*(4), Article e23219.

Haworth, B., Whittaker, J., & Bruce, E. (2016). Assessing the application and value of participatory mapping for community bushfire preparation. *Applied Geography, 76*, 115–127.

Food security and nutrition: Building a global narrative towards 2030. (2020). HLPE. HLPE Report No. 15.

EU food system monitoring dashboard. (2024). Joint Research Centre (JRC. https://datam.jrc.ec.europa.eu/datam/mashup/EU_FOOD_SYSTEM_MONITORING/index.html.

Jung, S., et al. (2024). Development of the Sustainable Diet Index–US. *Nutrition Journal, 23*, 15.

Levin, N., Lechner, A. M., & Brown, G. (2017). An evaluation of crowdsourced information for assessing the visitation and perceived importance of protected areas. *Applied geography, 79*, 115–126.

Lukyanenko, R., Parsons, J., Wiersma, Y. F., & Maddah, M. (2019). Expecting the unexpected: Effects of data collection design choices on the quality of crowdsourced user-generated content. *MIS Quarterly, 43*(2), 623–646. https://doi.org/10.25300/MISQ/2019/14198

Müller, A., Manners, R., Schumann, C., Mwizerwa, C., & Jimenez, D. (2024). *Addressing usability challenges in crowdsourced food system data*. AgriRxiv.

Manners, R., et al. (2022). Mobile phone-based diet quality surveillance in Rwanda. *Nutrients, 14*(12), Article 2456.

Mendoza, K., et al. (2024). Development of a crowdsourcing and gamification-based nutrition app in Mexico. *Scientific Reports, 14*(1), Article 6174.

Menozzi, D., Wongprawmas, R., Sogari, G., Gai, F., Parisi, G., & Mora, C. (2023). The role of objective and subjective knowledge on the attitude and intention of Italian consumers to purchase farmed and wild fish. *Agricultural and Food Economics, 11*(1), 47.

Mohammadi, A. (2025). Public participation in urban regeneration using XAI-driven decision support system. *Sustainable Cities and Society: Advances, 100013*.

Monteiro, C. A., et al. (2019). Ultra-processed foods: The NOVA classification. *Public Health Nutrition, 22*(5), 936–941.

Neta, S. P., et al. (2023). Review of sustainable diet indices: Dimensions, indicators, and gaps. *Nutrients, 15*(4), Article 912.

Oakden, L., Bridge, G., Armstrong, B., Reynolds, C., Wang, C., Panzone, L., & Serjeant, S. (2021). The importance of citizen scientists in the move towards sustainable diets. *Frontiers in Sustainable Food Systems, 5*, Article 596594.

Oliveira, R. (2022). FoodLink—A network for driving food transition in the Lisbon Metropolitan area. *Land, 11*(11), Article 2047.

Pertoldi, M., Fioretti, C., Guzzo, F., Testori, G., De Bruijn, M., Ferry, M., … Windisch, S. (2022). *Handbook of territorial and local development strategies (No. JRC130788)*. Joint Research Centre.

Polman, D., de Schutter, L., Begemann, S., Rivas, J. D. L., van Leeuwen, E., & Wensing, J. (2025). Cities in the loop: A social science perspective on the role of cities in food system circularity. *Regional Science Policy & Practice, 17*, Article 100238.

Polzin, S., Lusk, J. L., & Jaenicke, E. C. (2023). Measuring sustainable healthy diet behaviors: Development of a survey-based scale. *Food Quality and Preference, 104*, Article 104728.

Rei, L., et al. (2025). Multicriteria indicators of dietary sustainability: A systematic review. *Public Health Nutrition*.

Rocha, F., & Pires, S. M. (2025). Innovating in food systems: Challenges and opportunities in urban governance for sustainable transition. *Urban Governance, 5*, 466–481.

Rockström, J., Thilsted, S. H., Willett, W. C., Gordon, L. J., Herrero, M., Hicks, C. C., … DeClerck, F. (2025). The EAT–Lancet Commission on healthy, sustainable, and just food systems. *The Lancet, 406*(10512), 1625–1700.

Seconda, L., et al. (2020). Development and validation of an individual Sustainable Diet Index. *British Journal of Nutrition, 124*(10), 1063–1075.

Seok, D., et al. (2023). Can crowdsourced food images characterize restaurant nutrition environments? *Public Health Nutrition, 26*(3), 423–432.

Tepper, S., Geva, D., Shahar, D. R., Shepon, A., Mendelsohn, O., Golan, M., Adler, D., & Golan, R. (2021). The SHED Index. *European Journal of Nutrition, 60*, 3897–3909.

Thøgersen, J. (2025). Sustainable consumer behaviour. In *The routledge companion to marketing and sustainability* (pp. 185–199). Routledge.

Thomson, J. L., Landry, A. S., & Walls, T. I. (2023). Can United States adults accurately assess their diet quality? *American Journal of Health Promotion, 37*(4), 499–506.

Trijsburg, L., et al. (2020). Method for the development of WISH. *Nutrients, 13*(1), 93.

Willett, W., et al. (2019). Food in the anthropocene: The EAT–Lancet Commission. *The Lancet, 393*(10170), 447–492.

Worsley, A., et al. (2014). Nutrition knowledge and dietary behaviour. *Appetite, 76*, 134–143.